Chapter 2

Data Science Process

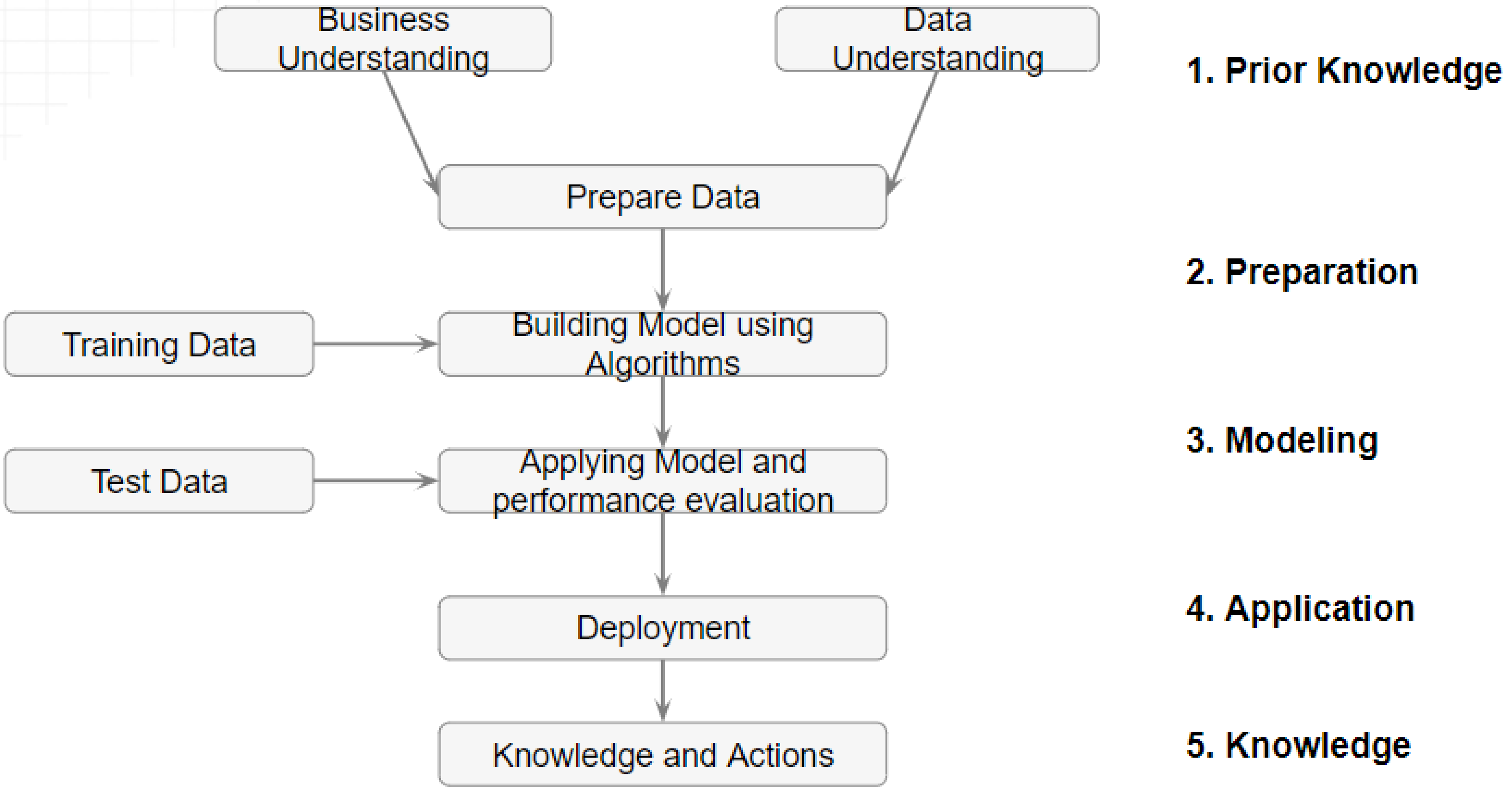
- **Data**
  - *a "given"* or *a fact* that represents something in real world
  - raw materials, can be processed, structured or unstructured
  - Data are elements of analysis
- **Information**
  - Data that have *meaning in context*
  - Data related
  - Data after manipulation
- **Knowledge**
  - In data science, knowledge refers to meaningful insights or patterns derived from data through analysis, modeling, and interpretation.
  - Acquired through experience or learning
  - It is a concept mainly for humans unlike data and information.

> - Knowledge is not information and information is not data.
> - Knowledge is derived from information in the same way information is derived from data.

Example

- **Data** – Raw, unprocessed facts (e.g., "500 movie reviews").
- **Information** – Processed or structured data (e.g., "300 positive and 200 negative movie reviews").
- **Knowledge** – Actionable insights extracted from the information (e.g., "Reviews mentioning 'amazing' and 'great' tend to be positive").
- **Wisdom –** Applying knowledge effectively for decision-making (e.g., "A recommendation system suggests movies based on positive reviews").

Data Science Process



Prior Knowledge

Gaining information on:

- Objective of the problem
- Subject area of the problem
- Data

Objective of the problem

The data science process starts with a need for analysis, a question, or a business objective. This is possibly the most important step in the data science Process. Without a well-defined statement of the problem, it is impossible to come up with the right dataset and pick the right data science algorithm.

Example : The business objective of this hypothetical case is: If the interest rate of past borrowers with a range of credit scores is known, can the interest rate for a new borrower be predicted?

Subject area of the problem

The process of data science uncovers hidden patterns in the dataset by exposing relationships between attributes. But the problem is that it uncovers a lot of patterns. The false or spurious signals are a major problem in the data science process. It is up to the practitioner to sift through the exposed patterns and accept the ones that are valid and relevant to the answer to the objective question. Hence, it is essential to know the subject matter, the context, and the business process generating the data.

Example: The lending business is one of the oldest, most prevalent, and complex of al the businesses. If the objective is to predict the lending interest rate, then it is important to know how the lending business works,

Data

Understanding how the data is collected, stored, transformed, reported, and used is essential to the data science process. There are quite a range of factors to consider: quality of the data, quantity of data. The objective of this step is to come up with a dataset to answer the business question through the data science process. For the following example, a sample dataset of ten data points with three attributes has been put together: identifier, credit score, and interest rate.

Data Types

- Two types of data: Labelled Data & Unlabelled Data
- **Labelled data**
- Specially designated attribute and the aim is to use the data given to predict the value of that attribute for instances that have not yet been seen. Data of this kind is called labelled.

Outlook=sunny	Temp=79	Humidity=88	Windy=false	Class=?

Outlook	Temp (°F)	Humidity (%)	Windy	Class
sunny	75	70	true	play
sunny	80	90	true	don't play
sunny	85	85	false	don't play
sunny	72	95	false	don't play
sunny	69	70	false	play
overcast	72	90	true	play
overcast	83	78	false	play
overcast	64	65	true	play
overcast	81	75	false	play
rain	71	80	true	don't play
rain	65	70	true	don't play
rain	75	80	false	play
rain	68	80	false	play
rain	70	96	false	play

Data Types

- **Unlabelled data**

  - Data that does not have any specially designated attribute is called unlabelled.
  - Here the aim is simply to extract the most information we can from the data available.

Age	Gender	Income	Profession	Tenure	City
35	M	60,000	IT	12	KRK
23	F	90,000	Sales	3	WAW
18	M	12,000	Student	1	KRK
42	F	128,000	Doctor	13	KRK
34	M	63,000	Manager	8	WAW
56	M	82,000	Teacher	30	WAW

Learning Methods

- **Supervised Learning**
- Data mining using labelled data is known as supervised learning.

- **Classification**
    - If the designated attribute is categorical, the task is called classification.
    - Classification is one form of prediction, where the value to be predicted is a label.
    - a hospital may want to classify medical patients into those who are at high, medium or low risk
    - we may wish to classify a student project as distinction, merit, pass or fail
    - Nearest Neighbour Matching, Classification Rules, Classification Tree, …

Learning Methods

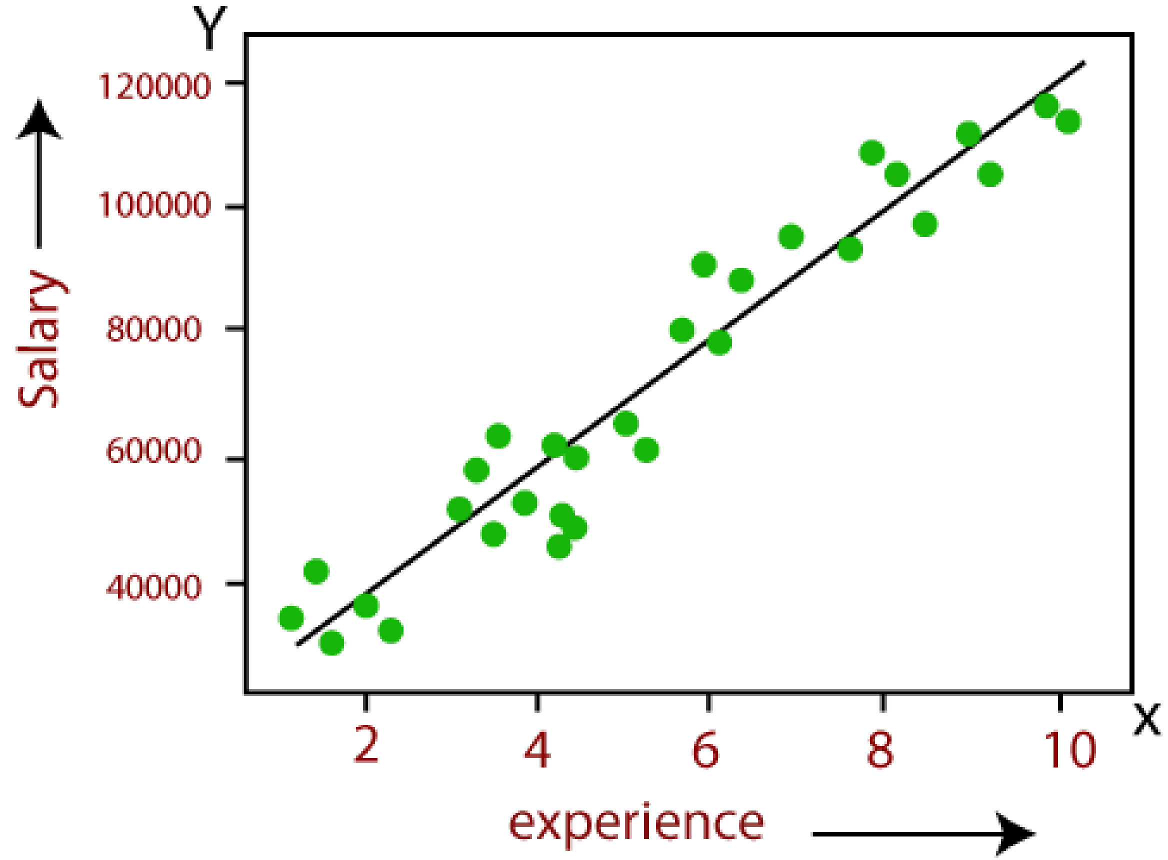
- **Numerical Prediction (Regression)**
  - If the designated attribute is numerical, the task is called numerical prediction (regression).
  - Numerical prediction (often called regression) is another. In this case we wish to predict a numerical value, such as a company's profits or a share price.

Example: Suppose there is a marketing company A, who does various advertisement every year and get sales on that. The list (right-side) shows the advertisement made by the company in the last 5 years and the corresponding sales. Now, the company wants to do the advertisement of $200 in the year 2019 and wants to know the prediction about the sales for this year.

Advertisement	Sales
$90	$1000
$120	$1300
$150	$1800
$100	$1200
$130	$1380
$200	??

Learning Methods

- **Regression (Numerical Prediction)**



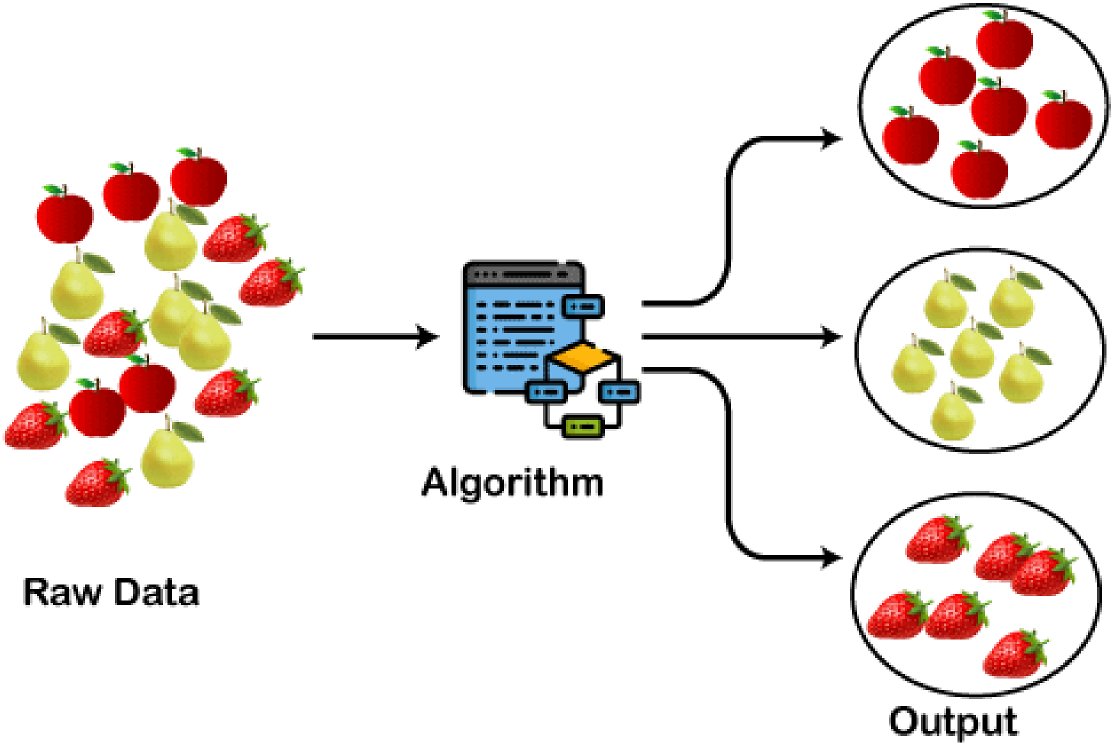
Learning Methods

- **Unsupervised Learning**
  - Data mining using unlabelled data is known as unsupervised learning.
  - **Association Rules**
    - Sometimes we wish to use a training set to find any relationship that exists amongst the values of variables, generally in the form of rules known as association rules.
    - APRIORI
    - Market Basket Analysis

Example, if a customer buys bread, he most likely can also buy butter, eggs, or milk, so these products are stored within a shelf or mostly nearby.
Possible **associations** include:
1.if customers purchase milk they also purchase bread **{milk}** → **{bread}**
2.if customers purchase milk and eggs they also purchase butter and bread **{milk, eggs}** → **{ butter, bread}**

Learning Methods

- **Clustering**
  - Clustering algorithms examine data to find groups of items that are similar.
  - K-Means Clustering, Agglomerative Hierarchical Clustering



- A dataset (example set) is a collection of data with a defined structure. Table 2.1 shows a dataset. It has a well-defined structure with 10 rows and 3 columns along with the column headers. This structure is also sometimes referred to as a “data frame”.

- A data point (record, object or example) is a single instance in the dataset. Each row in Table 2.1 is a data point. Each instance contains the same structure as the dataset.

- An attribute (feature, input, dimension, variable, or predictor) is a single property of the dataset. Each column in Table 2.1 is an attribute. Attributes can be numeric, categorical, date-time, text, or Boolean data types. In this example, both the credit score and the interest rate are numeric attributes.

Table 2.1 Data Set

Borrower ID	Credit Score	Interest Rate
01	500	7.31%
02	600	6.70%
03	700	5.95%
04	700	6.40%
05	800	5.40%
06	800	5.70%
07	750	5.90%
08	550	7.00%
09	650	6.50%
10	825	5.70%

- A label (class label, output, prediction, target, or response) is the special attribute to be predicted based on all the input attributes. In Table 2.1, the interest rate is the output variable.

Table 2.2 New Data With Unknown Interest Rate

Borrower ID	Credit Score	Interest Rate
11	625	?

Data Preparation

- Data Exploration
- Data quality
- Missing values
- Noisy values
- Invalid values
- Data types and Conversion
- Transformation
- Outliers
- Feature selection
- Sampling

Data Exploration

Data preparation starts with an in-depth exploration of the data and understanding the dataset better. Data exploration, also known as exploratory data analysis(EDA), provides simple tools to understand the data fully. Exploratory Data Analysis (EDA) is the process of analyzing and visualizing data to understand its structure, identify patterns, detect outliers, and gather insights before applying machine learning models .Data exploration approaches involve computing descriptive statistics and visualization of data.

Table 2.1 Data Set

Borrower ID	Credit Score	Interest Rate
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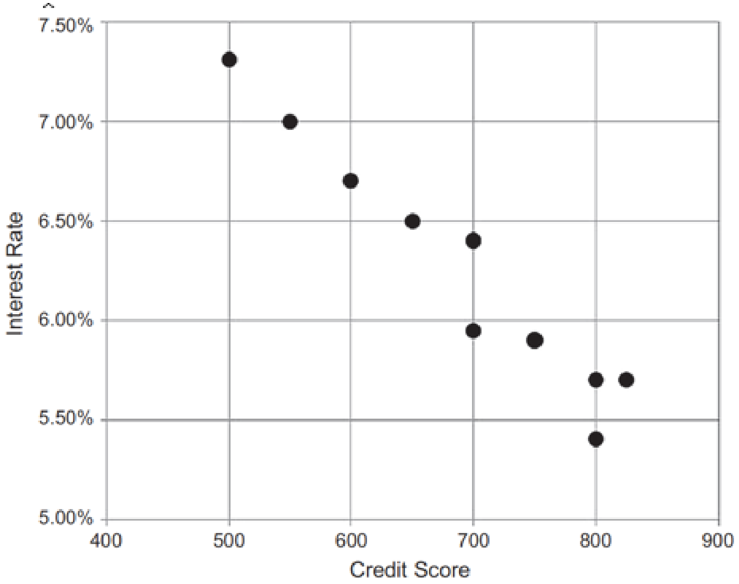


Data Exploration

Table 3.1 Iris Dataset and Descriptive Statistics (Fisher, 1936)

Observation	Sepal Length	Sepal Width	Petal Length	Petal Width
1	5.1	3.5	1.4	0.2
2	4.9	3.1	1.5	0.1
...	...	...	...	...
49	5	3.4	1.5	0.2
50	4.4	2.9	1.4	0.2
Statistics	**Sepal Length**	**Sepal Width**	**Petal Length**	**Petal Width**
Mean	5.006	3.418	1.464	0.244
Median	5.000	3.400	1.500	0.200
Mode	5.100	3.400	1.500	0.200
Range	1.500	2.100	0.900	0.500
Standard deviation	0.352	0.381	0.174	0.107
Variance	0.124	0.145	0.030	0.011

Data Quality

Data quality is the measure of how well suited a data set is to serve its specific purpose. Measures of data quality are based Data Correctness, Data Freshness and Data Completeness.

Data Correctness: How accurately the data value describes real-world facts. In other words, data correctness ensures that each data point is both accurate and reliable, meaning it reflects the truth as closely as possible. Data is accurate when it represents real-world values without error. For example, if a customer’s age is listed as 29 in a database, data accuracy would mean that this number actually reflects the customer's real age.. There are many potential root causes of collection issues such as collection noise, faulty data transformations, outdated data, or incorrect schema description.

Why Data Correctness is Important ?

Data correctness is critical because incorrect data can lead to poor decisions, unreliable results, and even damage to an organization’s credibility.

For example: In healthcare, incorrect patient data could lead to inappropriate treatment.

Data Freshness : This refers to how relevant the data is to describe the current state of an entity, and takes into consideration the timeliness of the data and how frequently it is updated. Data is correct when it is also up-to-date. Real-world values change over time, so data representing those values must be updated accordingly.

Example: A customer's address may change, and a database would need to reflect that change promptly to stay accurate.

Data Completeness: Data completeness refers to whether all necessary values are present in the dataset. For instance, if a dataset tracks employees, each employee entry should have essential information like name, ID, position, etc. Missing data can lead to an incomplete view of reality, which affects the correctness of the entire dataset.

Missing Values

In many real-world datasets data values are not recorded for all attributes. This can happen simply because there are some attributes that are not applicable for some instances, a malfunction of the equipment used to record the data, a data collection form to which additional fields were added after some data had been collected, information that could not be obtained, e.g. about a hospital patient

k-nearest neighbor (k-NN) algorithm for classification tasks are often robust with missing values. Neural network models for classification tasks do not perform well with missing attributes, and thus, the data preparation step is essential for developing neural network models

SoftEng	ARIN	HCI	CSA	Project	Class
A	B	A	B	B	Second
A	B		B	B	Second
B		A		A	Second

Methods to Handle Missing Values

- **Discard Instances**
- **Replace by Most Frequent/Average Value**

Discard Instances

- This is the simplest strategy: delete all instances where there is at least one missing value and use the remainder.
- It has the advantage of avoiding introducing any data errors. Its disadvantage is that discarding data may damage the reliability of the results derived from the data

Replace by Most Frequent/Average Value

- A less cautious strategy is to estimate each of the missing values using the values that are present in the dataset.
- A straightforward but effective way of doing this for a categorical attribute is to use its most frequently occurring (non-missing) value
- In the case of continuous attributes, it is likely that no specific numerical value will occur more than a small number of times. In this case the estimate used is generally the average value.

Noisy Values

- A noisy value to mean one that is valid for the dataset, but is incorrectly recorded
- The number 69.72 may accidentally be entered as 6.972, or a categorical attribute value such as brown may accidentally be recorded as another of the possible values, such as blue.

Invalid Values

- 69.7X for 6.972 or bbrown for brown
- An invalid value can easily be detected and either corrected or rejected

Data types and Conversion

The attributes in a dataset can be of different types, such as continuous numeric (interest rate), integer numeric (credit score), or categorical. For example, the credit score can be expressed as categorical values (poor, good, excellent) or numeric score. Different data science algorithms impose different restrictions on the attribute data types.

Sample Employee Dataset

Employee ID	Gender	Age	Marital Status	Income (USD)
E001	Male	28	Yes	45000
E002	Female	34	No	52000
E003	Male	40	Yes	67000
E004	Female	29	No	48000
E005	Male	45	Yes	89000
E006	Female	32	Yes	72000
E007	Male	50	No	94000
E008	Female	27	No	41000
E009	Male	38	Yes	76000
E010	Female	42	Yes	81000

Transformation

In some data science algorithms like k-NN, the input attributes are expected to be numeric and normalized, because the algorithm compares the values of different attributes and calculates distance between the data points. Normalization prevents one attribute dominating the distance results because of large values. To overcome this problem ,we generally normalize the values of continuous attributes.

The idea is to make the values of each attribute run from 0 to 1. In general, if the lowest value of attribute A is min and the highest value is max, we convert each value of A, say a, to (a − min)/(max − min).

Mileage (miles)	Number of doors	Age (years)	Number of owners
18,457	2	12	8
26,292	4	3	1

Transformation

YearsExperience	Salary
1.1	39343
1.3	46205
1.5	37731
2	43525
2.2	39891
2.9	56642
3	60150
3.2	54445
3.2	64445
3.7	57189
3.9	63218
4	55794
4	56957

First 13 rows of the salary data

Normalized Value

Original Value

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Maximum Value of x

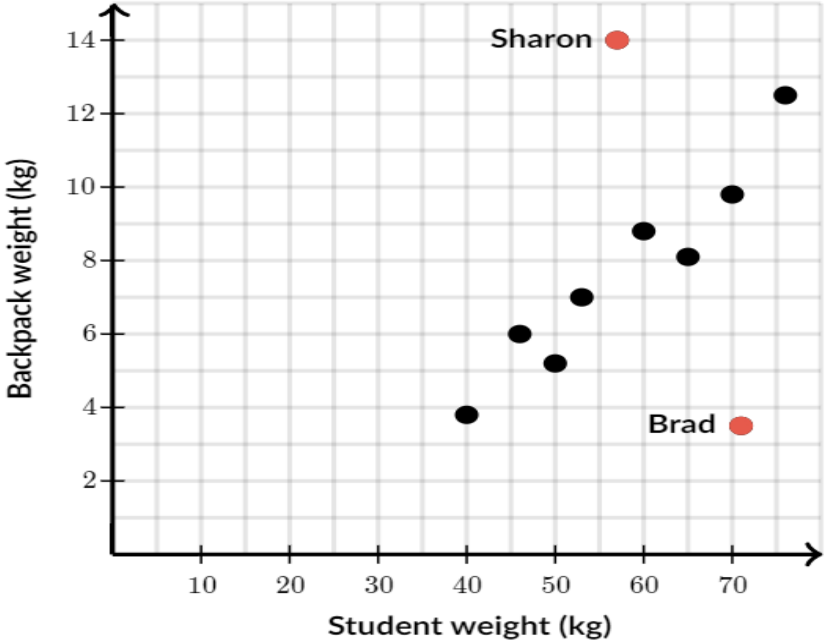
Minimum Value of x

```
##    YearsExperience     Salary
## 1       0.00000000 0.01904087
## 2       0.02127660 0.10009450
## 3       0.04255319 0.00000000
## 4       0.09574468 0.06843846
## 5       0.11702128 0.02551382
## 6       0.19148936 0.22337586
```

First 6 rows of Normalized data (Min-Max Normalization)

Outliers

Outliers are those data points that are significantly different from the rest of the dataset. They are often abnormal observations that skew the data distribution, and arise due to inconsistent data entry, or erroneous observations. Detecting outliers may be the primary purpose of some data science applications, like fake email detection, fraud or intrusion detection.



Feature Selection

Reducing the number of attributes, without significant loss in the performance of the model, is called feature selection.

many data science problems involve a dataset with hundreds to thousands of attributes. In text mining applications, every distinct word in a document forms a distinct attribute in the dataset. Not all the attributes are equally important or useful in predicting the target. The presence of some attributes might be counter productive. Some of the attributes may be highly correlated with each other, like annual income and taxes paid. A large number of attributes in the dataset significantly increases the complexity of a model and may degrade the performance of the model due to the curse of dimensionality

Glaucoma Dataset (62 features)

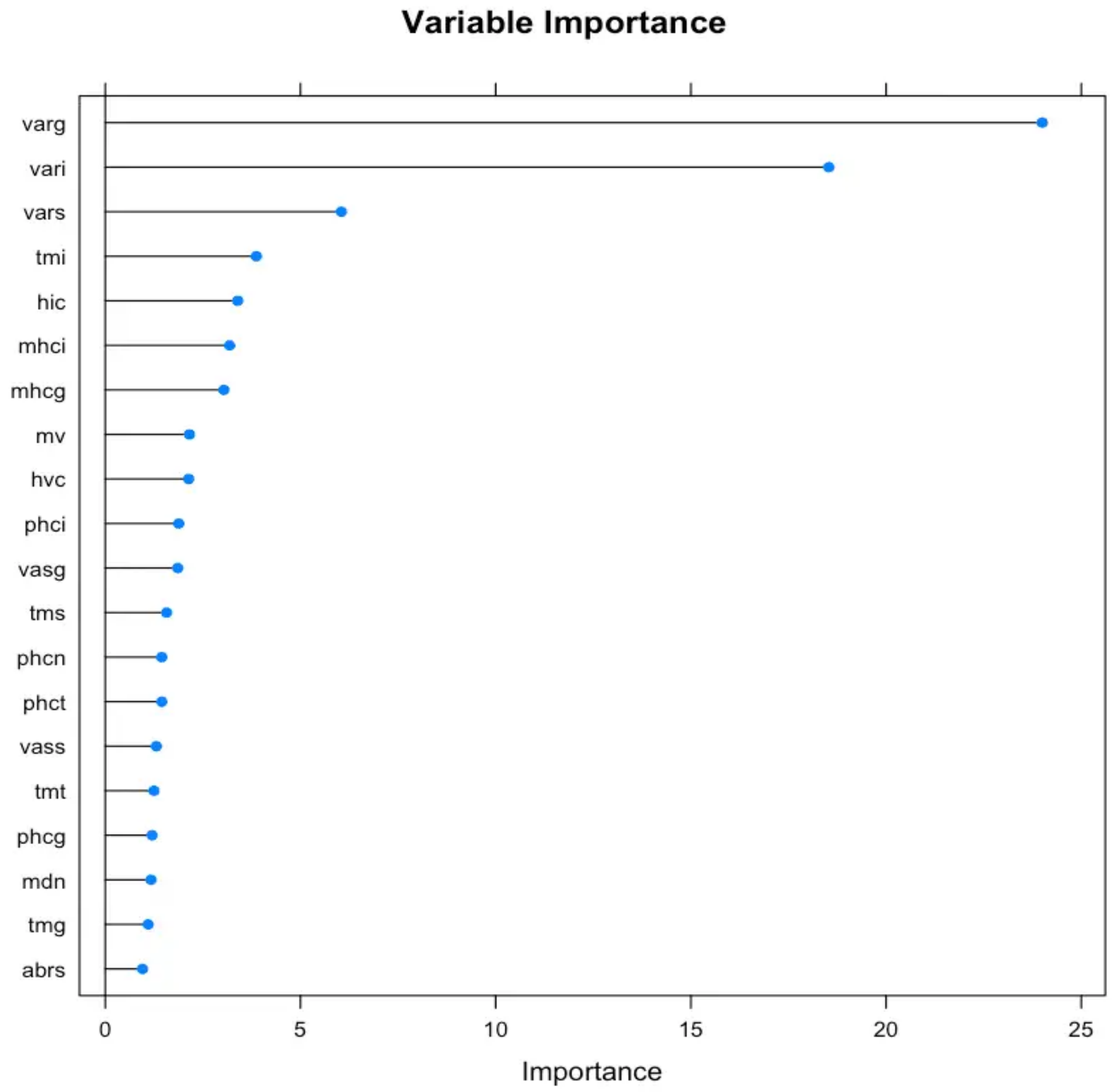
Feature Selection

	ag	at	as	an	ai	eag	eat	eas	ean	eai	...	tmt	tms	tmn	tmi	mr	rnf	mdic	emd	mv	Class
2	2.220	0.354	0.580	0.686	0.601	1.267	0.336	0.346	0.255	0.331	...	-0.018	-0.230	-0.510	-0.158	0.841	0.410	0.137	0.239	0.035	normal
43	2.681	0.475	0.672	0.868	0.667	2.053	0.440	0.520	0.639	0.454	...	-0.014	-0.165	-0.317	-0.192	0.924	0.256	0.252	0.329	0.022	normal
25	1.979	0.343	0.508	0.624	0.504	1.200	0.299	0.396	0.259	0.246	...	-0.097	-0.235	-0.337	-0.020	0.795	0.378	0.152	0.250	0.029	normal
65	1.747	0.269	0.476	0.525	0.476	0.612	0.147	0.017	0.044	0.405	...	-0.035	-0.449	-0.217	-0.091	0.746	0.200	0.027	0.078	0.023	normal
70	2.990	0.599	0.686	1.039	0.667	2.513	0.543	0.607	0.871	0.492	...	-0.105	0.084	-0.012	-0.054	0.977	0.193	0.297	0.354	0.034	normal
16	2.917	0.483	0.763	0.901	0.770	2.200	0.462	0.637	0.504	0.597	...	0.087	0.018	-0.094	-0.051	0.965	0.339	0.333	0.442	0.028	normal

Glaucoma Dataset (62 features)

Feature Selection

Regularized Random Forest – Variable Importance

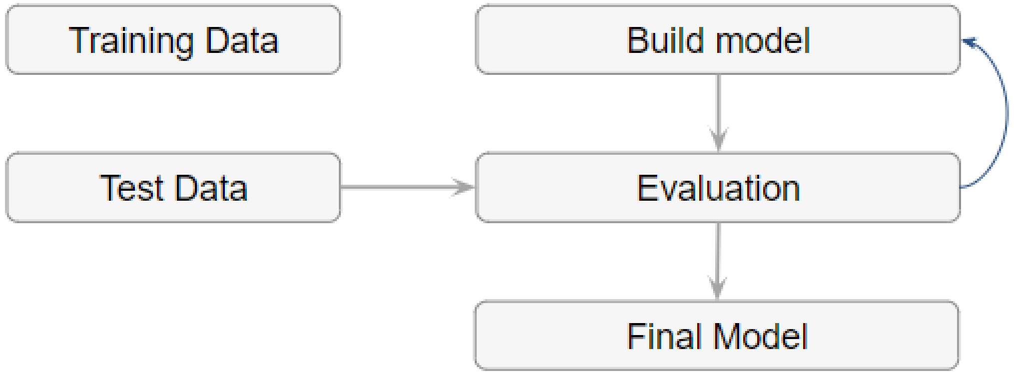


Data Sampling

Sampling is a process of selecting a subset of records as a representation of the original dataset for use in data analysis or modeling. The sample data serve as a representative of the original dataset with similar properties, such as a similar mean. Sampling reduces the amount of data that need to be processed and speeds up the build process of the modeling. In most cases, to gain insights, extract the information, and to build representative predictive models it is sufficient to work with samples. Theoretically, the error introduced by sampling impacts the relevancy of the model, but their benefits far outweigh the risks.

MODELING

A model is the abstract representation of the data and the relationships in a given dataset.



MODELING

Splitting Training and Test data sets: The modeling step creates a representative model inferred from the data. The dataset used to create the model, with known attributes and target, is called the training dataset.

The validity of the created model will also need to be checked with another known dataset called the test dataset or validation dataset. To facilitate this process, the overall known dataset can be split into a training dataset and a test dataset. A standard rule of thumb is two-thirds of the data are to be used as training and one-third as a test dataset

Training Set and Test Set

Table 2.3 Training Data Set

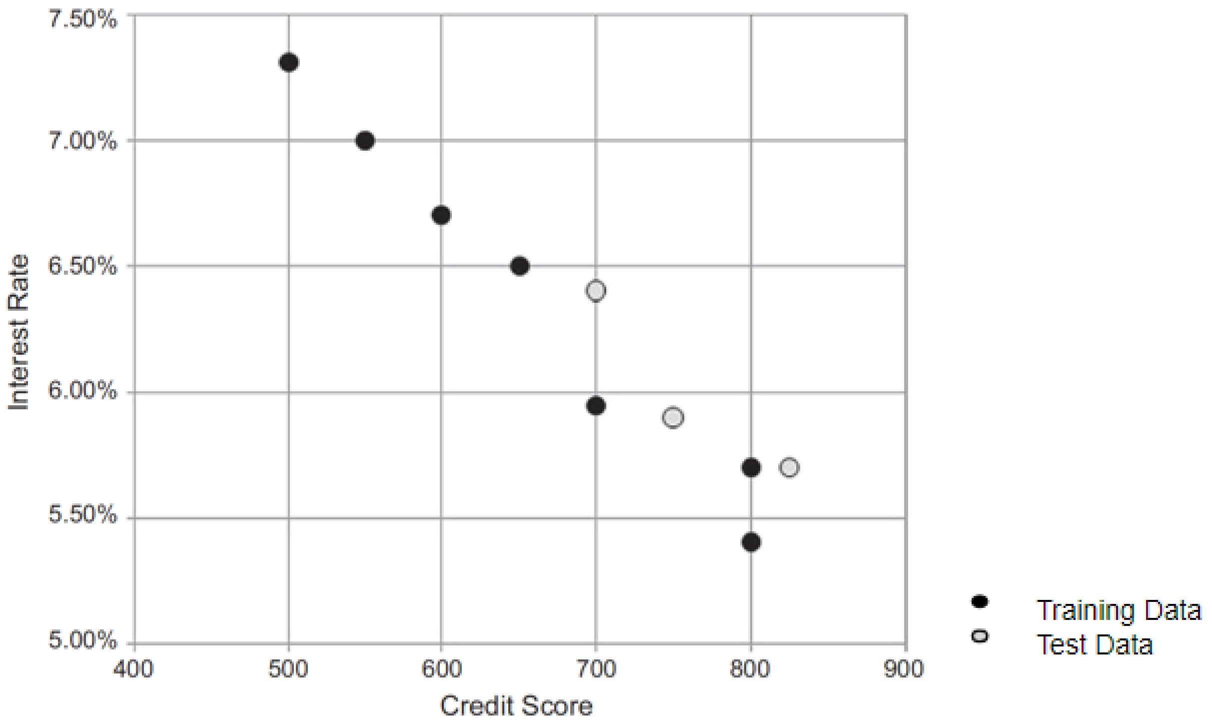
Borrower	Credit Score (X)	Interest Rate (Y)
01	500	7.31%
02	600	6.70%
03	700	5.95%
05	800	5.40%
06	800	5.70%
08	550	7.00%
09	650	6.50%

Table 2.4 Test Data Set

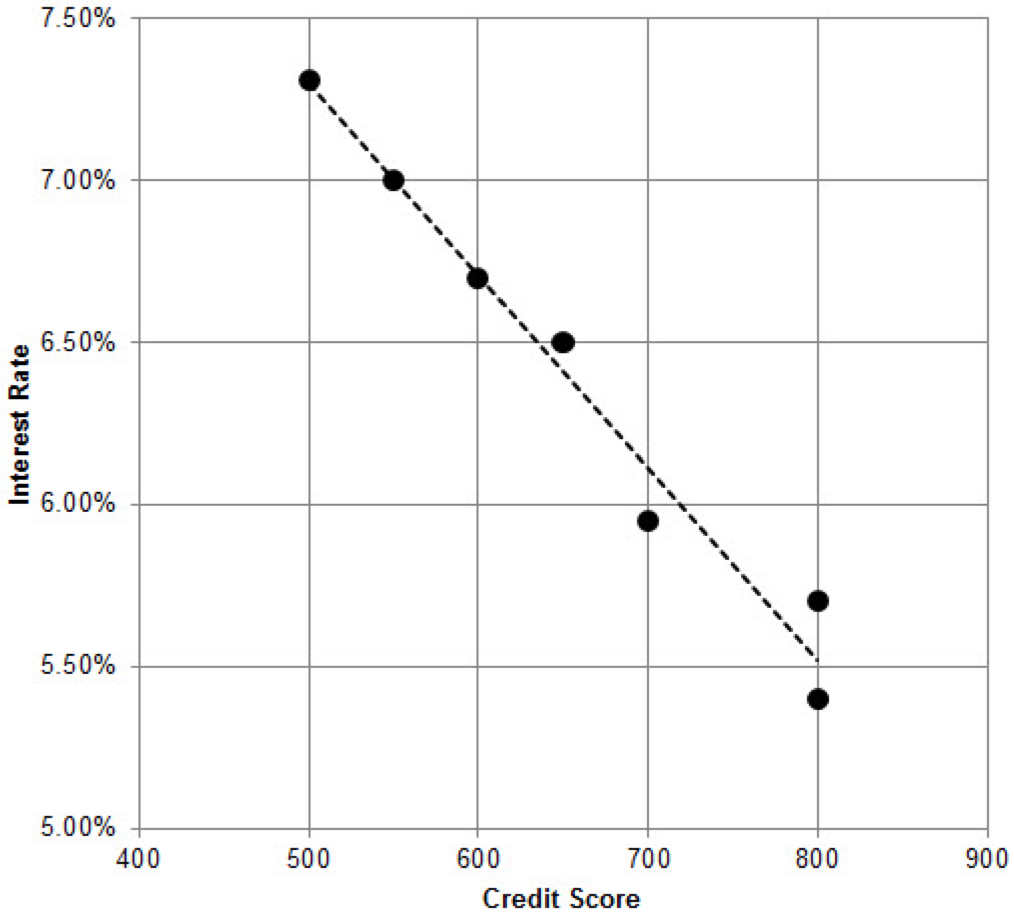
Borrower	Credit Score (X)	Interest Rate (Y)
04	700	6.40%
07	750	5.90%
10	825	5.70%

MODELING

Splitting Training and Test data sets



MODELING



7.50%
7.00%
6.50%
6.00%
5.50%
5.00%
Interest Rate
400
500
600
700
800
900
Credit Score

MODELING

Evaluation of test dataset

Table 2.5 Evaluation of Test Data Set

Borrower	Credit Score (X)	Interest Rate (Y)	Model Predicted (Y)	Model Error
04	700	6.40%	6.11%	-0.29%
07	750	5.90%	5.81%	-0.09%
10	825	5.70%	5.37%	-0.33%

Application

Product readiness
Technical integration
Model response time
Remodeling
Assimilation

Knowledge

Posterior knowledge